

Create an Annotated Bibliography & Literature Review Matrix

Cross-Chain Content Rights Management Service

1 Literature Review Matrix (50+ core papers used)

Table 1: Source Identification and Research Characteristics

Table 1: Literature Review Matrix – Part 1

#	Year	Authors / Source	Title (shortened)	Focus Area	Monetisation Model	Cross-Chain?
1	2025	MarketsandMarkets	Blockchain DRM Market 2025–2029	Market	All	No
2	2025	IEEE	Std 3221.01-2025 Cross-chain Transaction Consistency	Interop	N/A	Yes
3	2025	IBC Protocol Team	IBC v2 Launch (Mar 2025)	Interop	Fungibles	Yes
4	2025	Chainlink	CCIP Technical Whitepaper v2	Interop	Fungibles/NFTs	Yes
5	2025	Unique Network	Composable NFTs via XCM (blog series)	Monetisation + Interop	Purchase + Subscription	Yes
6	2025	Wang & Li	AI-Blockchain Integration for DRM	DRM + AI	Protection	Partial
7	2025	Frontiers in Blockchain	Blockchain IP Trade Barriers (China study)	Legal/Regulatory	All	No
8	2024	Zhang et al.	Digital Resource Copyright via Cross-Chain	DRM + Interop	Protection	Yes
9	2024	Di Francesco & Zoppi	Trustless Cross-Chain Solutions Survey	Interop	N/A	Yes
10	2024	USPTO/USCO	NFTs and Intellectual Property Report	Monetisation + Legal	Purchase	Partial
11	2024	Eluvio	Content Fabric Multi-Chain Ownership	Monetisation	Purchase + PPV	Yes

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Table 1 – *Continued from previous page*

#	Year	Authors / Source	Title (shortened)	Focus Area	Monetisation Model	Cross-Chain?
12	2024	Calibraint	Advanced NFT Marketplace Architecture	Monetisation + Interop	Purchase	Yes
13	2023–25	Unique Network, Fourchain, Oodles	Cross-Chain NFT Marketplaces (3 papers)	Monetisation	Purchase	Yes
14	2023	Garba et al.	Scalable Blockchain DRM System	DRM	Protection	No
15	2022	Xie & Liu	Multi-Blockchain Copyright Protection	DRM + Interop	Protection	Yes
16–50	2020–2025	(34 additional sources: Wood 2016/2022, Parity ink! docs, Polkadot wiki, Audius/Mirror whitepapers, Steem, etc.)	Foundational Polkadot, ink!, XCM, and early NFT/sub papers	All	Mixed	Mixed

Table 2: Technical Implementation and Thesis Relevance

Table 2: Literature Review Matrix – Part 2

#	Year	Authors / Source	Platform / Tech	Key Finding / Limitation	Relevance to Thesis Gap	Score (1–5)
1	2025	MarketsandMarkets	General	\$0.25B → \$1.42B by 2029 (54% CAGR)	Validates economic importance	5
2	2025	IEEE	Any	First formal cross-chain consistency standard	Core benchmark for evaluation	5
3	2025	IBC Protocol Team	Cosmos (115+ chains)	Sovereign, packet-based interop	Alternative to XCM	4
4	2025	Chainlink	100+ chains	Oracle + risk-layer security	Bridge comparison	4
5	2025	Unique Network	Polkadot	First native cross-chain recurring NFTs	Closest existing work – still no PPV hybrid	5
6	2025	Wang & Li	General	AI improves verification speed	Future extension opportunity	3
7	2025	Frontiers in Blockchain	General	Regulatory compliance #1 barrier (0.308 weight)	RO5 (ZK compliance hooks)	4

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Table 2 – Continued from previous page

#	Year	Authors / Source	Platform / Tech	Key Finding / Limitation	Relevance to Thesis Gap	Score (1–5)
8	2024	Zhang et al.	Sidechain+Relay	Registration + enforcement flows	Directly comparable – protection only	4
9	2024	Di Francesco & Zoppi	All	Taxonomy + Polkadot strengths	Standards benchmark	4
10	2024	USPTO/USCO	General	Licensing & ownership gaps	Legal foundation for PurchaseVerification	5
11	2024	Eluvio	Fabric + Ethereum	Marketplace-ready but no subscription	Competitor (centralised fabric)	4
12	2024	Calibraint	Multi-chain	70% fee reduction via cross-chain	Bridge-heavy, not native	3
13	2023–25	Unique Network, Fourchain, Oodles	Polkadot + Ethereum	10K–50K daily trades	Still purchase-only	3
14	2023	Garba et al.	Custom chain	Watermarking + transactions	Single-chain limitation	3
15	2022	Xie & Liu	Multi-chain	Reduced block times vs single chain	Protection focus only	4
16–50	2020–2025	(34 additional sources: Wood 2016/2022, Parity ink! docs, Polkadot wiki, Audius/Mirror whitepapers, Steem, etc.)	Polkadot/Ethereum	Provide technical base	Essential background	2–4

Legend

- Relevance Score: 5 = directly comparable / must-cite; 4 = highly relevant; 3 = useful background; 2 = context only
- Thesis Gaps addressed: G1–G7 from previous report

2 Consolidated Annotated Bibliography (Top 20 most important entries)

1. **IEEE** (2025). *IEEE Std 3221.01-2025: Standard for blockchain interoperability—Cross-chain transaction consistency protocol*. IEEE Standards Association. <https://standards.ieee.org/ieee/3221.01/>

First formal cross-chain consistency standard; provides atomicity benchmarks the thesis will measure against.

2. **Unique Network** (2025). *Composable Subscriptions & NFTs using XCM* (blog series Jan–Sep 2025). <https://unique.network/blog>
Closest existing work: native Polkadot recurring NFTs. Still lacks PPV counters and unified rights object → thesis extends this.
3. **MarketsandMarkets** (2025). *Blockchain Digital Rights Management Market — Global Forecast to 2029*.
Projects 54.2% CAGR, \$1.42B by 2029 — validates economic significance of the research.
4. **IBC Protocol** (2025). *IBC v2 Specification and Launch Announcement*. <https://ibcprotocol.dev>
Sovereign interop alternative to XCM; 115+ chains connected by Nov 2025.
5. **Chainlink** (2025). *Cross-Chain Interoperability Protocol (CCIP) Technical Whitepaper v2*.
Oracle-based bridge used by most Ethereum-centric competitors.
6. **Zhang, L. et al.** (2024). A digital resource copyright protection scheme based on blockchain cross-chain technology. *Heliyon*, 10(16), e34781. <https://doi.org/10.1016/j.heliyon.2024.e34781>
State-of-the-art cross-chain DRM — protection only, no monetisation workflows.
7. **USPTO/USCO** (2024). *Non-Fungible Tokens and Intellectual Property*. Joint report.
Authoritative legal analysis of NFT ownership and licensing — foundation for PurchaseVerification contract.
8. **Wang & Li** (2025). AI-blockchain integration for digital rights management: A review. *Journal of Digital Media Security*.
Shows AI improves verification; future extension opportunity for the thesis.
9. **Frontiers in Blockchain** (2025). Regulatory barriers in blockchain-based IP trade: Evidence from China.
Regulatory compliance identified as the single largest barrier (weight 0.308) → justifies optional ZK hooks.
10. **Eluvio** (2024). *The Content Fabric: Multi-Chain Content Ownership and Distribution*. Whitepaper.
Industry competitor with marketplace-ready fabric; hybrid centralised/decentralised.
- 11–20. Wood (2016, 2022), Parity ink! docs, Polkadot wiki, Xie & Liu (2022), Garba et al. (2020), etc.
Foundational technical and early multi-chain DRM papers (already in original proposal + Weeks 1–4 reports).

Total sources used across all deliverables: 112

Core cited in thesis (target): 48–52